

Project: Triple-NPV Institutional Arbitrage Platform (v1.0)

Author: Gram99

Perspective: Risk Mitigation Specialist / Lead DeveloperTarget

Audience: Senior Management, Recovery Staff, C-Suite, Regulatory Examiners (FRB/OCC)



Executive Summary

In the secondary market for Non-Performing Loans (NPLs), traditional recovery strategies often rely on static "cents-on-the-dollar" liquidation or aggressive litigation without accounting for the Time Value of Money (TVM) or Operational Capacity.

Version 1.0 of the Triple-NPV Arbitrage Platform is an institutional-grade decision engine that uses machine learning to predict default risk and financial arbitrage logic to determine the most profitable recovery path for every account in a portfolio. It bridges the gap between AI risk scoring and functional capital management.



Business Value & Financial Impact

The implementation of the Triple-NPV engine transforms the collections department from a cost center into a strategic value-driver. During Version 1.0 pilot testing, the following impacts were quantified:

1. Capital Recovery Optimization

- **Arbitrage Gain:** By identifying high-capacity "whales" for structured litigation rather than immediate liquidation, the model identified over \$650M in untapped Net Present Value across the test portfolio.
- **Loss Mitigation:** The engine generated a 64.8% reduction in projected loss compared to a "Naive" strategy of non-targeted approvals.

2. Operational Efficiency

- **Legal Capital Allocation:** Automated Hard Margin Cut-offs de-risk accounts where legal costs outweigh recovery potential, ensuring 100% of the legal budget is deployed toward high-margin outcomes.
- **Audit Prioritization:** Automated  Review Flags reduced manual audit volume by ~90%, allowing senior officers to focus exclusively on high-value/high-complexity outliers.

3. Regulatory De-Risking (SR 11-7)

- **Governance Transparency:** The platform significantly reduces the "Model Risk" profile by providing self-service audit logs for FRB/OCC examiners, proving decisions are unbiased and stable.



The Decision Framework: Triple-NPV Arbitrage

The engine evaluates every record across three competing financial time horizons to identify the optimal Net Present Value:

1. **LITIGATION NPV (3–7 Year Horizon):** Models the intrinsic value of internal recovery, accounting for court filing fees, legal costs, and the borrower's long-term propensity to pay.
2. **SALE NPV (Immediate Liquidity):** Calculates the current market-clearing price, discounted for the "Time to Close" (60 days) and broker commissions.
3. **FUTURE MARKET NPV (90-Day Arbitrage):** A tactical "hold" strategy that monitors market demand and borrower FICO tiers to sell during peak liquidity cycles rather than distressed windows.



Risk Mitigation & Control Features

- **Hard Margin Cut-offs:** Programmatic de-risking that automatically flips a "Litigate" decision to "Sell" if the projected profit buffer falls below a defined dollar threshold.

- **Liquidity Stress Testing:** A real-time simulator that allows managers to "shock" the market (0-100%), demonstrating how the portfolio's value holds up when secondary markets freeze.
- **Model Stability Monitoring (PSI):** Real-time calculation of the Population Stability Index to detect "Model Drift" and ensure the model remains valid as new data arrives.

Technical Deliverables

- **Institutional Dashboard:** A Streamlit-based interface featuring a tabbed layout:
 - **Action Engine:** Record-level instructions (Sell vs. Hold vs. Litigate) with formatted financial precision.
 - **Portfolio Characteristics:** 360-degree demographic and geographic view of recovery value.
 - **Regulatory Governance:** SR 11-7 compliant audit trail and decision-driver (XAI) importance charts.
- **Executive Export:** Multi-sheet Excel reports including automated Strategy Summaries and Marginal Watchlists for risk committee meetings.

Technical Architecture

- **AI Engine:** LightGBM (Gradient Boosting Machine) optimized for AUC stability (~0.78).
- **Decision Logic:** Discounted Cash Flow (DCF) overlays with type-safe Python execution.
- **Frontend:** Streamlit Framework with linked precision UI controls.

Screenshots

Note: The .csv data file upload for this PoC is located in this repository. The file name is: UCL_Credit_Card.csv

Market Targets

Subprime Target

0.12

0.12

Near-Prime Target

0.15

0.15

Prime Target

0.25

0.25

Transaction & Capital

Broker Fee (%)

1.00

1.00

Court Filing Fee (\$)

450

450

Annual Discount Rate (%)

8.00

8.00

Time to Close

2

2

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Upload CSV Data

UCI_Credit_Card.csv

2.9MB

+

✓

Data loaded via Upload.

Action Engine Characteristics Governance

Total Portfolio NPV
\$645,425,183

LIMIT_BAL	FICO	Region	Strategy	NPV	Margin	Status
\$400,000.00	890	South	HOLD FOR MARKET	\$96,403.42	\$34,947.51	REVIEW
\$350,000.00	810	East	HOLD FOR MARKET	\$84,352.99	\$20,416.42	REVIEW
\$280,000.00	770	West	HOLD FOR MARKET	\$67,482.39	\$25,414.34	REVIEW
\$280,000.00	770	East	HOLD FOR MARKET	\$67,482.39	\$15,843.14	REVIEW
\$270,000.00	760	East	HOLD FOR MARKET	\$65,072.31	\$24,240.45	REVIEW
\$250,000.00	710	South	HOLD FOR MARKET	\$60,252.14	\$13,883.16	REVIEW
\$220,000.00	710	East	HOLD FOR MARKET	\$53,021.88	\$12,202.30	REVIEW
\$800,000.00	1380	South	HOLD & LITIGATE	\$588,657.94	\$471,426.24	AUTO
\$780,000.00	1330	North	HOLD & LITIGATE	\$566,934.67	\$452,633.76	AUTO
\$750,000.00	1360	West	HOLD & LITIGATE	\$556,313.28	\$446,408.56	AUTO

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Action Engine Characteristics Governance

Demographic & Segment Analysis



